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## EXPERIMENT NO:2

### AIM

To Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

### Program

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
file_path = 'sales-data-new.csv'
df = pd.read_csv(file_path)
print(df.head())

print(df.isnull().sum())

df['Sales'] = df['Sales'].fillna(df['Sales'].mean())
df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)
print(df.describe())
product_summary = df.groupby('Product').agg({
    'Sales': 'sum',
    'Quantity': 'sum'
}).reset_index()

print(product_summary)
plt.figure(figsize=(10, 6))
plt.bar(product_summary['Product'], product_summary['Sales'])
plt.xlabel('Product')
```

```
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.show()
df['Date'] = pd.to_datetime(df['Date'], dayfirst=True)
sales_over_time = df.groupby('Date').agg({
    'Sales': 'sum'
}).reset_index()
```

```
plt.figure(figsize=(10, 6))
plt.plot(sales_over_time['Date'], sales_over_time['Sales'])
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('Sales Over Time')
plt.xticks(rotation=45)
plt.show()
pivot_table = df.pivot_table(
    values='Sales',
    index='Region',
    columns='Product',
    aggfunc='sum',
    fill_value=0
)
```

```
print(pivot_table)
```

```
correlation_matrix = df[['Sales', 'Quantity']].corr()
```

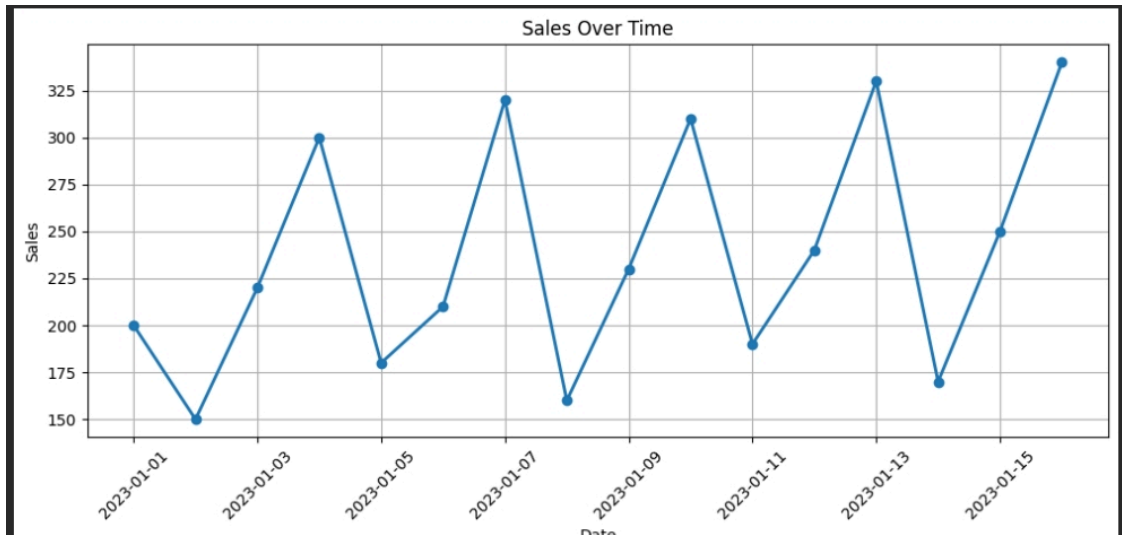
```
print(correlation_matrix)
plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()
```

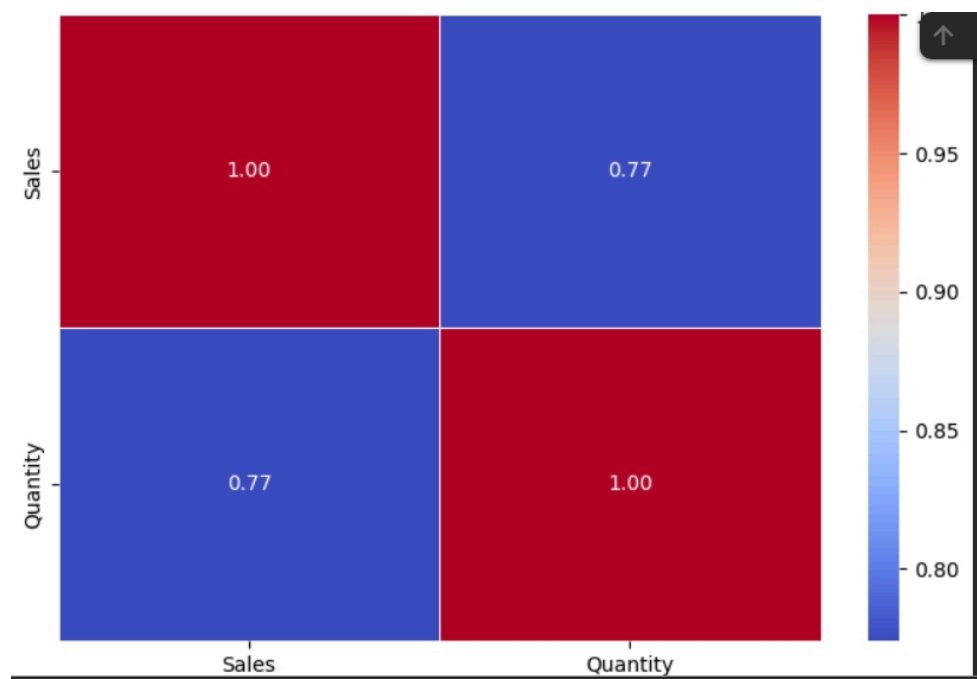
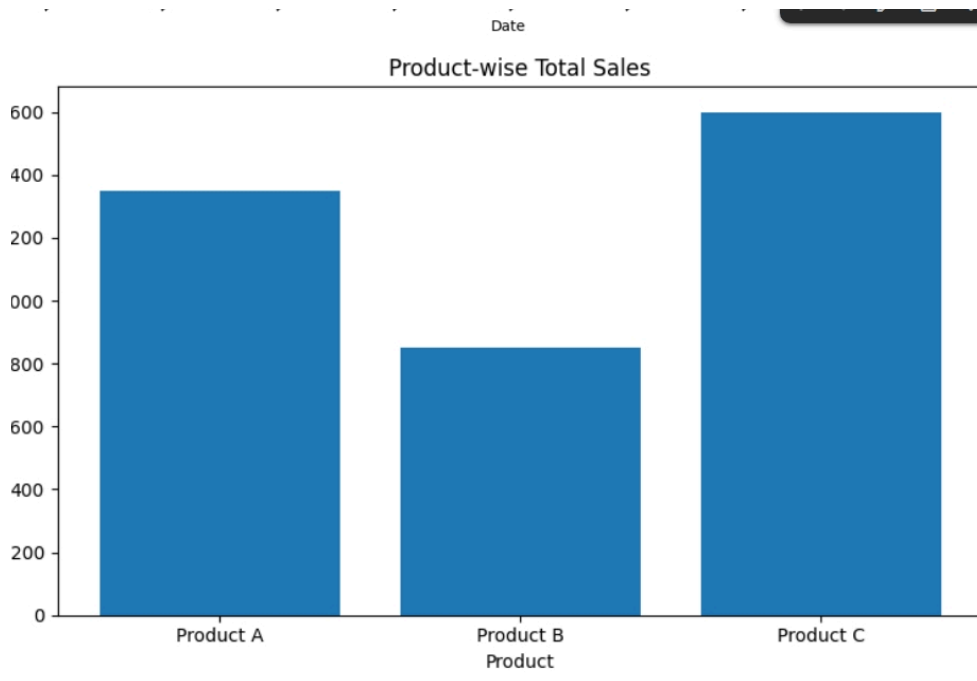
Output

```

      Date  Product  Sales  Quantity  Region
0  01-01-2023  Product A    200         4  North
1  02-01-2023  Product B    150         3  South
2  03-01-2023  Product A    220         5  North
3  04-01-2023  Product C    300         6  East
4  05-01-2023  Product B    180         4  West
Date      0
Product    0
Sales      0
Quantity   0
Region     0
dtype: int64

      Sales  Quantity
count  16.000000  16.000000
mean    237.500000   5.375000
std     64.031242   1.746425
min    150.000000   3.000000
25%    187.500000   4.000000
50%    225.000000   5.500000
75%    302.500000   7.000000
max    340.000000   8.000000
      Product  Sales  Quantity
0  Product A   1350        33
1  Product B    850        17
2  Product C   1600        36
```





## RESULT

Thus the sales dataset was successfully analyzed using pandas & the required statistical summary product-wise analysis, line graph, bar graph, correlation matrix, and heatmap were successfully generated.